Chemistry 3A – Spring 2026 Name _____

Sections 19320-19321

Special Midterm Examination #1 ID _____

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which number below is standard (decimal) notation for 4.10×10^{-5}?
   a) 410,000 b) 41,000 c) 0.0000410 d) 0.000041

2. Of the orders shown below from largest to smallest size (atomic radius), which show the correct order?
   a) Ca > Si > Cs > F b) Cs > Ca > Si > F c) F > Si > Ca > Cs d) F > Ca > Si > Cs

3. What is the correct electronic configuration for element **sulfur**?
   a) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$ **b)** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ **c)** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ **d)** $1s^2\ 2s^2\ 2p^6\ 3s^1\ 3p^5$

4. Which element has seven (7) valence electrons?
   a) Nitrogen b) Oxygen c) Fluorine d) Neon

5. The elements in the ***d***-block are known by what name?
   a) Metalloids b) Alkaline Earth Metals c) Transition Metals d) None of a, b, or c

6. Which of these ions has the same electron configuration as **neon** (Ne)?
   a) Na^- b) F^+ c) O^{2-} d) Mg^{2-}

7. Using the electron filling rules (Aufbau principle, how many orbitals of unpaired electrons will there be for an atom of the element **scandium** (**Sc**)?
   a) 3 b) 2 c) 1 d) 0

8. What is the result of the division operation with correct significant digits: $20.6807 \div 6.91$
   a) 2.99287 b) 2.9929 c) 3.0 d) 2.99

9. What is the density of a marble with 8.1 gram mass placed inside a graduated cylinder having a volume of water reading 10.0 mL before placing the marble and reading 13.7 mL after the marble is placed? The values of all choices have units of grams/milliliter (g/mL)
   a) 2.19 b) 0.81 c) 29.97 d) 0.46

10. Consider the following atomic symbols, where X is any known element:
    I. ${}^{75}_{33}X$, II. ${}^{72}_{30}X$, III. ${}^{74}_{32}X$, IV. ${}^{74}_{30}X$. Which pairs are isotopes of the same element?
    a) I and III b) I and IV c) II and IV d) II and III

11. Which subatomic particle(s) are in a cloud outside the nucleus?
    a) electrons only b) protons only c) protons and electrons d) protons and neutrons

12. Which statement is part of Dalton's atomic theory?
    a) Atoms can be divided into protons and neutrons.
    b) Electrons orbit the nucleus in fixed paths.
    c) Atoms of the same element have identical properties.
    d) Atoms of different elements have the same mass.

13. A researcher records a mass of 450.0 grams. How should this be expressed in scientific notation?
    a) 4.5×10^{-2} g b) 4.5×10^{2} g c) 4.500×10^{-2} g d) 4.500×10^{2} g

14. Which principle states that electrons should be filled unpaired (single electron in an orbital) with the same spin direction before they are paired in orbitals with opposite spin direction?
a) Hund's rule b) Pauli exclusion principle c) Aufbau principle d) Dalton's rule

15. If a substance has a temperature of 273 K, which temperature in Celsius is closest to it?
a) 27°C b) 573°C c) 30°C d) 0°C

16. Which step of the scientific method involves using controlled conditions usually to determine changes of a single variable and testing validity of a hypothesis?
a) observation b) law c) theory d) experiment

17. Perform the calculation and apply decimal-place rules: 14.2009 + 0.035 + 2.81
a) 17.045 b) 17.1 c) 17.05 d) 17.0

18. Which of the following elements is considered a **non-metal**?
a) carbon b) bismuth c) niobium d) potassium

19. In Bohr's model, electrons moving from higher to lower energy levels:
a) emit a photon b) absorb energy c) become ions d) change atomic number Z

20. Perform the calculation and apply significant-digit rules: 5.60 × 0.030
a) 0.17 b) 0.168 c) 0.2 d) 0.170

21. Which statement related with protons is correct?
a) They have a mass of about 1 amu b) They orbit the nucleus outside of it
c) Sum of protons + electrons equals mass number ***A*** d) They have a negative charge

22. How many orbitals does the ***p*** orbital type have?
a) 1 b) 3 c) 5 d) 7

23. If the density of an object was multiplied by the volume of the object, what units would be used or most appropriate for the result?
a) grams (g) b) milliliters (mL) c) atomic mass units (amu) d) centimeters (cm)

24. How many neutrons does the beryllium-9 isotope, which has a Z = 4 value, have?
a) 4 b) 5 c) 9 d) 13

25. The energy levels of electrons in orbit around nucleus are quantized. What does this mean?
a) Electrons can move between levels using any amount of energy ("continuous" change)
b) Electrons can change their charge when moving between levels
c) Electrons convert from matter into pure energy when changing levels
d) Electrons can move between energy levels only with specific amounts of energy

Continue to next page for True/False questions

26. The atomic mass value of an element in the Periodic Table is the sum of the all the masses of the isotopes (in amu) times the natural abundance (or fraction) of that isotope in the universe.
    a) true b) false

27. In the scientific method, the start of that method begins with observations.
    a) true b) false

28. Ionization energy is the energy required for the electron of an atom to be set free from the atom.
    a) true b) false

29. The identity of an element is determined by its Z value, the number of protons.
    a) true b) false

30. Neutrons have much smaller mass than protons.
    a) true b) false

31. None of the zero digits in the number 0.0004940 are significant.
    a) true b) false

32. In scientific notation, the significand (coefficient) must be a number between 1 and 10
    a) true b) false

33. When the electron in the hydrogen (H) atom moves from the n = 1 energy level to the n = 2 energy level, that electron is in the "ground state".
    a) true b) false

34. Water becomes steam at 0K.
    a) true b) false

35. As electrons are added one-by-one to the elements of the 4th period, they will fill up the **4***s* orbitals and then start filling the **3*d*** orbitals.
    a) true b) false

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